

One Softmax Head in Four Restores Ceiling Recall: A Pre-Registered Replication with a Censored Effect and an Unreachable Frontier

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Abstract

Linear-attention architectures trade the quadratic cost of softmax attention for a fixed-size recurrent state, and pay for it with a capacity ceiling on associative recall. Hybridizing the two along the head axis is an established remedy. We report a pre-registered replication that quantifies it, and two results that the pre-registration forced us to state against our own interest. First, in a multi-query associative recall benchmark at $d_{\text{model}} = 64$ with four heads, the delta-rule condition falls to .921 and .820 accuracy at loads 96 and 128, while every hybrid condition sits at 1.000; the paired difference at the pre-registered evaluation load is $+.0792$, 95% CI $[+.0747, +.0838]$, four times the pre-registered margin, positive in eight of eight seeds. Second, the minimum dose tested—one softmax head in four—already restores the ceiling (worst cell deficit 1.22×10^{-4}), so the claim about the two-head mixture is one of *sufficiency, not necessity*, and the cost of the remedy is bounded above at one quarter of the heads. Critically, the effect is *censored*: all hybrid conditions saturate, so $+.0792$ is a lower bound and the measured statement is about the delta rule—*it falls below ceiling at loads where hybrids do not*—not about how much better hybrids are. A pre-registered calibration then searched for a regime with dynamic range, and failed under its own rules: doubling the key pool and the load left the reference condition at 1.000, so the frontier is unreachable at this scale and the equivalence prediction it would have served is recorded as *not run*, never as confirmed. We argue the durable contribution is the pre-registration itself, and report the ledger of conveniences it refused.

1 Introduction

Attention with a softmax kernel compares every query against every key, which costs $O(n^2)$ in sequence length but places no architectural bound on how many associations a layer can retrieve. Linear attention replaces the comparison with a fixed-size recurrent state updated online, which costs $O(n)$ and does place such a bound: the state has a finite number of slots, and when the number of stored associations approaches that number, retrieval degrades. This trade-off is well established, and so is the remedy of mixing the two mechanisms within a layer (Arora et al., 2023; 2024; Yang et al., 2024). It has since been demonstrated at scale: Tan et al. (2026) hybridize full and linear attention along the head axis and report large gains at long context.

This paper does not claim the mechanism as novel. It was already known when our protocol was frozen, and Tan et al. (2026) appeared a month before it. What we contribute is a measurement of it under rules written in advance, which is rare in this literature, and three consequences of taking those rules seriously:

1. The remedy saturates at the smallest dose we tested—*one head in four*—so the usual two-and-two mixture is sufficient but not necessary, and the price of the fix is bounded above at a quarter of the heads.

2. The headline number is *censored by a ceiling*, which changes what it means. We report the corrected statement rather than the flattering one.
3. A pre-registered attempt to escape the ceiling failed under its own criteria, and the prediction it would have served is recorded as not run.

2 Method

2.1 Pre-registration

The protocol was frozen on 2026-07-22 (SHA-256 2f8ebb82...) and publicly deposited before any comparison condition was trained. A follow-up pre-registration for the comparison reported here was frozen on 2026-07-23 (SHA-256 0b93a36f...) with a signed tag and a server-side timestamp preceding every run it governs. Two later amendments—one adding the minimum-dose condition, one specifying the regime search—were each frozen, hashed, and publicly anchored before the run they authorize. A gate script re-hashes all frozen artifacts and refuses to launch if any differs.

2.2 Task and conditions

The benchmark is multi-query associative recall: a sequence presents L key-value pairs followed by queries, and the model must return the value bound to each queried key. Load L is the number of stored pairs; a pool of N_K keys supplies distractors that never appear. Accuracy is top-1 exactness on the queried positions.

All conditions share one architecture (193,493 parameters; $d_{\text{model}} = 64$, four heads, RMSNorm per head) and differ only in which heads use which mechanism: **softmax** (four softmax heads), **delta** (four delta-rule heads), **mix22** (two and two), and **mix13** (one softmax head and three delta-rule heads). Parameter counts are identical by construction; FLOPs are not, and cannot be—softmax is $O(n^2)$ and the delta rule is $O(n)$ —so the pre-registration equalizes parameters, reports FLOPs, and declares the direction of the bias: the hybrids run at a *compute disadvantage* relative to pure softmax, which makes any confirmation conservative.

2.3 Margins, seeds, and the evaluation load

Eight seeds per condition. The equivalence margin is not a fixed number but a rule fixed in advance: $R_{11} = \max(.02, 1.5 \times \text{SD across seeds of the reference})$, instantiated and recorded before any comparison condition was inspected; here it evaluates to .0200. Differences are paired by seed. The evaluation load is likewise defined by rule—the smallest load at which the reference falls below 95%—which yields $L = 96$.

A three-verdict scheme is used throughout: *difference*, *equivalence*, or *inconclusive*. Neither equivalence nor difference is ever declared by default.

3 Results

3.1 The delta rule falls below ceiling; every hybrid does not

Table 1 gives accuracy by load at the common budget.

Table 1: Top-1 accuracy by load, eight seeds, common budget $N = 10,000$.

condition	$L8$	$L16$	$L32$	$L64$	$L96$	$L128$
delta	1.000	1.000	1.000	.985	.921	.820
softmax	1.000	1.000	1.000	1.000	1.000	1.000
mix22	1.000	1.000	1.000	1.000	1.000	1.000
mix13	1.000	1.000	1.000	1.000	1.000	1.000

The delta-rule condition is the only one that leaves the ceiling, and it does so monotonically with load. At the pre-registered evaluation load the paired difference between the two-and-two mixture and the delta rule is $+0.0792$, 95% CI $[+0.0747, +0.0838]$, which is $4.0\times$ the pre-registered margin of $.0200$ and positive in eight of eight seeds. The prediction is confirmed in both the primary and the secondary table, and the pre-registered discordance rule—which would have downgraded the verdict had the two tables disagreed—did not fire. An independent re-implementation of the statistic, written without importing any of the analysis code and using only the standard library, reproduces the mean exactly and the confidence interval within Monte Carlo noise.

3.2 One head in four is enough

The minimum-dose condition was added by a frozen amendment carrying an interpretation table written *before* the run: each possible outcome was mapped in advance to the claim it would license. The outcome fell in the row where the hybrid restores the ceiling at the smallest dose tested. Its worst cell across the entire grid is a deficit of 1.22×10^{-4} from unity—within the $R_{11}/10$ tolerance fixed for “at ceiling”—and its paired difference against the delta rule at $L96$ is $+0.0792$, 95% CI $[+0.0748, +0.0837]$, again eight of eight seeds.

Two consequences follow, and the frozen table states both:

- The claim about `mix22` becomes one of **sufficiency, not necessity**. Two softmax heads restore the ceiling, but so does one; the experiment does not license “two are needed.” The cost of the remedy is bounded above at a quarter of the heads.
- **No dose-response curve is obtained**. All three hybridized points saturate, so the monotonicity prediction cannot be evaluated. This is reported as not evaluable, not as confirmed.

3.3 What the number actually means: censoring

It is tempting to report $+0.0792$ as “the hybrid outperforms the delta rule by eight points.” That reading is wrong, and the pre-registration is what caught it.

Every hybrid condition sits at the ceiling across the whole grid—the worst cell of `mix22` anywhere is $.999512$. Its capacity is therefore not measured but **censored**: we observe that it has not yet failed, not how far it is from failing. An instrument that saturates cannot report the magnitude of what exceeds it.

The measured statement is consequently about the *delta rule*:

The delta rule alone falls below ceiling at loads where hybrid conditions do not.

and $+0.0792$ is a **lower bound** on a difference whose true value is unknown and may be much larger. This is a weaker claim about the hybrid than the raw number suggests, and a sharper one about the linear-attention baseline. It is also, we note, a different claim from those in the scaled hybrid literature, which report end-task gains rather than a bound on where a pure linear-attention baseline first fails.

3.4 Secondary predictions

The pre-registered descriptive statistic placing the hybrid between the two pure conditions gives $f = (C_3 - C_2)/(C_1 - C_2) = 1.001$: the hybrid is exactly at the softmax ceiling, which is the censoring above restated.

The prediction about where degradation begins is confirmed in all three of its parts: onset at median $L_0 = 64$; strict monotonicity of degradation with load ($\rho = -1.000$); and an accelerating rather than linear decay ($+0.0361$, 95% CI $[+0.0335, +0.0390]$).

The prediction of an anticorrelation between capacity and correctness is **not conclusive** and is reported as such. Its intended test was not evaluable at the evaluation load (the reference SD fell below the pre-registered usability threshold), so the pre-specified fallback applied, which makes the comparison cross-load; the resulting correlation is $-.455$ with a confidence interval that crosses zero. A diagnostic pre-registered for exactly this purpose rules out the stopping-point confound as the explanation, leaving statistical power as the limitation. With eight seeds the study cannot resolve it.

4 The frontier is not reachable at this scale

The censoring above has an obvious remedy: find a regime where the reference has dynamic range, and re-run the comparison there. A rule for choosing such a regime was frozen before any calibration data existed. It specifies a target band for reference accuracy— $[.50, .80]$, where the upper edge is $1 - 10R_{11}$ rather than a matter of taste—an ordered, enumerated search space of two regimes, a compute budget split in two declared halves, and four exhaustive outcome branches.

Both rungs were run. The first ($N_K = 256$, $L \leq 128$) returned 1.000 at every load. The second ($N_K = 512$, $L \leq 256$)—double the key pool and double the load, retrained from scratch—returned the values in Table 2: still at ceiling everywhere, worst cell .9999.

Table 2: Reference accuracy on the second calibration rung ($N_K = 512$).

L	8	16	32	64	96	128	192	256
acc@1	1.000	1.000	.9999	1.000	1.000	1.000	1.000	.9999

With both rungs closed and neither meeting the band, the frozen decision procedure emits its third branch: the frontier lies beyond what this budget can measure at this d_{model} . The consequences were written in advance and are mechanical. The equivalence prediction that required a non-saturated regime is recorded as **not run—never as confirmed**, on the explicit ground that running it in a saturated regime yields a row that cannot fail. Reducing d_{model} is routed to a future campaign as documented redesign, because shrinking it breaks the “same architecture” invariant that makes campaigns comparable.

We stress the asymmetry that makes this reportable rather than embarrassing. The branch was not chosen after seeing the result; it was one of four written before any calibration data existed, together with the rule that distinguishes it from “we ran out of budget”—a distinction that mattered, because an earlier reading of the same evidence would have declared the frontier unreachable while one rung remained unrun. Running it cost seventeen minutes of a T4.

An exploratory search outside the frozen space (varying d_{model} , reported here only to document why it is not pursued) found the regime degenerate rather than merely difficult: at $d_{\text{model}} = 8$ each head writes into a state of four numbers, and the hybrid condition becomes *unstable rather than incapable*—it peaks near .52 and then collapses. A comparison there would measure numerical collapse, not the hypothesis.

5 The ledger of conveniences

If the durable contribution of this work is the pre-registration rather than the mechanism, then the honest way to report it is to publish what the rules refused, not only what they allowed. We list the decisions where the frozen protocol ran against our interest.

- **A tempting headline was downgraded.** “The hybrid beats the delta rule by eight points” became “the delta rule falls where hybrids do not, by at least eight points,” because the hybrid was censored.
- **A claim of necessity became one of sufficiency** when the minimum dose worked, under an interpretation table written before the run.
- **A prediction was retired rather than reported.** A stopping rule we had proposed would have halted training while the model was still improving; it was withdrawn and replaced by a power-based rule, and the replacement was declared non-retroactive.
- **An amendment that would have exempted a condition from extension was rewritten** after an audit found its threshold excluded the very condition it was written for.
- **An inconclusive result stayed inconclusive.** The capacity–correctability prediction could have been reported as a trend; its interval crosses zero, so it is reported as underpowered.

- **A defense of a safeguard was withdrawn** when it was found to compare a between-regime displacement against within-regime noise, and replaced by a measured one.

Two further measurements are worth recording because they bear on how such comparisons should be reported. First, an internal stopping criterion—frozen with a public timestamp before data collection—was later shown to be undetectable in principle at the achieved signal-to-noise ratio (Speranza, 2026). Second, changing the compute backend (CPU versus a T4 GPU), with seed, data, and code held fixed, moves validation accuracy by roughly 30% of the decision margin in the late training regime. Studies that report “mean $\pm$ SD over N seeds” omit a term of comparable order.

6 Limitations

The censoring is the central one: the magnitude of the hybrid advantage is not measured, only bounded below, and the calibration established that no accessible regime removes the ceiling on this task at this scale. The result therefore transfers to the two other tasks in the programme only as a hypothesis—the frozen rule that governs the regime search says so explicitly, since those tasks carry their own chance level and difficulty.

The anticorrelation prediction is underpowered at eight seeds. The common training budget is declared rather than derived: the delta-rule condition had not converged at the pre-registered cap in three of eight seeds, so it is reported as a fixed declared budget, which biases against that condition and is therefore conservative for the confirmed prediction. Finally, this is a small-scale study ($\sim 10^5$ parameters, synthetic task); it quantifies a mechanism under controlled conditions and does not establish end-task gains at scale, for which the reader should consult the scaled literature cited above.

Broader Impact Statement

This work studies the capacity of small associative-recall models on a synthetic benchmark. We see no direct ethical concerns arising from the experiments themselves. The methodological argument—that pre-registration changes what a reported effect is allowed to claim—is intended to make results of this kind harder to overstate, which we regard as its main contribution.

Open Practices

All artifacts needed to reproduce every number in this paper—the frozen protocol and its amendments, raw per-seed results, the analysis code, an independent re-implementation of the headline statistic, and the calibration records—are publicly available: they are mirrored, for review, in an anonymized copy at <https://anonymous.4open.science/r/hybrid-heads-mqar-2072>. Each pre-registered artifact carries the SHA-256 hash quoted in Section 2.1 above, a signed tag, and a server-side timestamp that precedes the run it governs; the hashes can be recomputed from the repository copy and matched against the values printed in this paper. The non-anonymous public anchors—a DOI deposit and a public repository whose signed tags carry the timestamps—exist and predate every run reported here; they are withheld here only because they identify the author, and will be disclosed on acceptance. The analysis scripts run on CPU.

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